

Research paper

Activity-dependent extracellular potassium changes in unmyelinated *versus* myelinated areas in olfactory regions of the isolated female guinea-pig brain

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ARTICLE INFO

Keywords:

Myelin
Extracellular potassium
Potassium buffering
Hyperexcitability
Piriform cortex

ABSTRACT

The potassium released in the extracellular space during neuronal activity is rapidly removed by glia and neurons to maintain tissue homeostasis. Oligodendrocyte-derived myelin axonal coating contributes to potassium buffering and is therefore crucial to control brain excitability.

We studied activity-dependent extracellular potassium ($[K^+]_o$) changes in the piriform cortex (PC), a region that features highly segregated bundles of myelinated and unmyelinated fibers. Four-aminopyridine (4AP; 50 μ M) treatment or patterned high-frequency stimulations (hfST) were utilized to generate $[K^+]_o$ changes measured with potassium-sensitive electrodes in the myelinated lateral olfactory tract (LOT), in the unmyelinated PC layer I and in the myelinated deep PC layers in the *ex vivo* isolated guinea-pig brain.

Seizure-like events induced by 4AP are initiated by the abrupt $[K^+]_o$ rise in the layer I formed by unmyelinated fibers (Uva et al., 2017). Larger $[K^+]_o$ shifts occurred in unmyelinated layers compared to the myelinated LOT. LOT hfST that mimics pre-seizure discharges also generated higher $[K^+]_o$ changes in unmyelinated PC layer I than in LOT and deep PC layers. The treatment with the Kir4.1 potassium channel blocker BaCl₂ (100 μ M) enhanced the $[K^+]_o$ changes generated by hfST in myelinated structures.

Our data show that activity-dependent $[K^+]_o$ changes are intrinsically different in myelinated vs unmyelinated cortical regions. The larger $[K^+]_o$ shifts generated in unmyelinated structures may represent a vehicle for seizure generation.

1. Introduction

Focal seizure activity is sustained by complex interactions between synaptic and non-synaptic events/processes that are still incompletely understood. Experimental research largely focuses on the role of excitatory and inhibitory synaptic interactions for the interpretation of seizure generation and progression (Devinsky et al., 2018; Wenzel et al., 2023). The possibility that focal seizures are boosted by local changes in activity-dependent extracellular ions that involve both synaptic and non-synaptic mechanisms should be considered. The association between seizure onset and increases in extracellular potassium ($[K^+]_o$) has been consistently and extensively reported (reviewed in Fröhlich et al., 2008; de Curtis et al., 2018). Early pioneering studies performed with

electrodes selectively sensitive to extracellular K^+ concentration demonstrated a correlation between epileptiform events and $[K^+]_o$ rises in different animal models and on *in vitro* slices treated with epileptogenic drugs (Fertziger and Ranck, 1970; Prince et al., 1973; Moody et al., 1974; Heinemann et al., 1977, 1986; Pumain and Heinemann, 1985; Avoli, 1996; Avoli et al., 1996).

We have recently described a novel seizure-like pattern induced by the K^+ channel blocker 4-aminopyridine (4AP) in the piriform cortex (PC) maintained *in vitro* (Carriero et al., 2010; Uva et al., 2013, 2017; de Curtis et al., 2021); both in the isolated guinea-pig brain and in tangential PC slices, 4AP-induced seizure-like event (SLE) features a rapid increase of the $[K^+]_o$ in the absence of an obvious and excessive neuronal firing (Uva et al., 2013, 2017). The PC has a peculiar laminar

Abbreviations: 4AP, 4-aminopyridine; $[K^+]_o$, extracellular potassium concentration; hfST, high-frequency stimuli trains; LOT, lateral olfactory tract; PC, piriform cortex; SLEs, seizure-like events.

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<https://doi.org/10.1016/j.expneurol.2024.114884>

Received 15 April 2024; Received in revised form 26 June 2024; Accepted 8 July 2024

Available online 9 July 2024

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structure characterized by a thick superficial layer I formed by densely arranged unmyelinated fibers that lose their myelin coating upon leaving the lateral olfactory tract (LOT; for review see [Neville and Haberly, 2004](#)). We hypothesized ([Uva et al., 2017](#)) that the rapid $[K^+]_o$ rise observed during PC SLEs could be due to the release of K^+ from depolarized unmyelinated axons of superficial PC layer I ([Scholfield, 1990](#)), in concert with a saturation of the $[K^+]_o$ buffering competence of the glia ([Grimaldi et al., 2001](#)).

The aim of the present study is to define the differences in the release and accumulation of K^+ in layers of the PC that are selectively rich of either unmyelinated or myelinated fibers. We compared the $[K^+]_o$ rises induced by epileptiform discharges i) in the LOT formed by the highly myelinated axons of olfactory bulb mitral cells, ii) in superficial layer I composed by LOT fibers that lose their myelin coating entering the PC and iii) in PC layer II-III containing myelinated axons ([Neville and Haberly, 2004](#); see also [Fig. 1A](#)). Finally, we interfered with $[K^+]_o$ buffering dynamics by applying the Kir4.1 channel blocker BaCl₂ ([Bay](#)

and [Butt, 2012](#); [Larsen et al., 2016](#)) and monitored the changes in $[K^+]_o$ in superficial and deeper layers of PC. We demonstrate that K^+ containment in the extracellular space is more efficient in brain tissue that contains myelinated fibers compared to unmyelinated regions.

2. Materials and methods

2.1. Isolated guinea-pig brain maintained in vitro

Adult Hartley female guinea-pigs (150–200 g; Charles River, Calco, Italy) were utilized for the experiments. Animals were deeply anesthetized with isoflurane (Aerrane, Baxter, 4%). Surgery started with the exposition of the heart; intracardiac perfusion was performed with cold (4 °C), carboxygenated (95%O₂/5%CO₂) solution containing 126 mM NaCl, 3 mM KCl, 1.2 mM KH₂PO₄, 1.3 mM MgSO₄, 2.4 mM CaCl₂, 26 mM NaHCO₃, 15 mM glucose, 2.1 mM HEPES and 3% Hydroxyethyl starch (Voluven M.W. 130 KDa, Fresenius Kabi, Italy) at pH 7.1

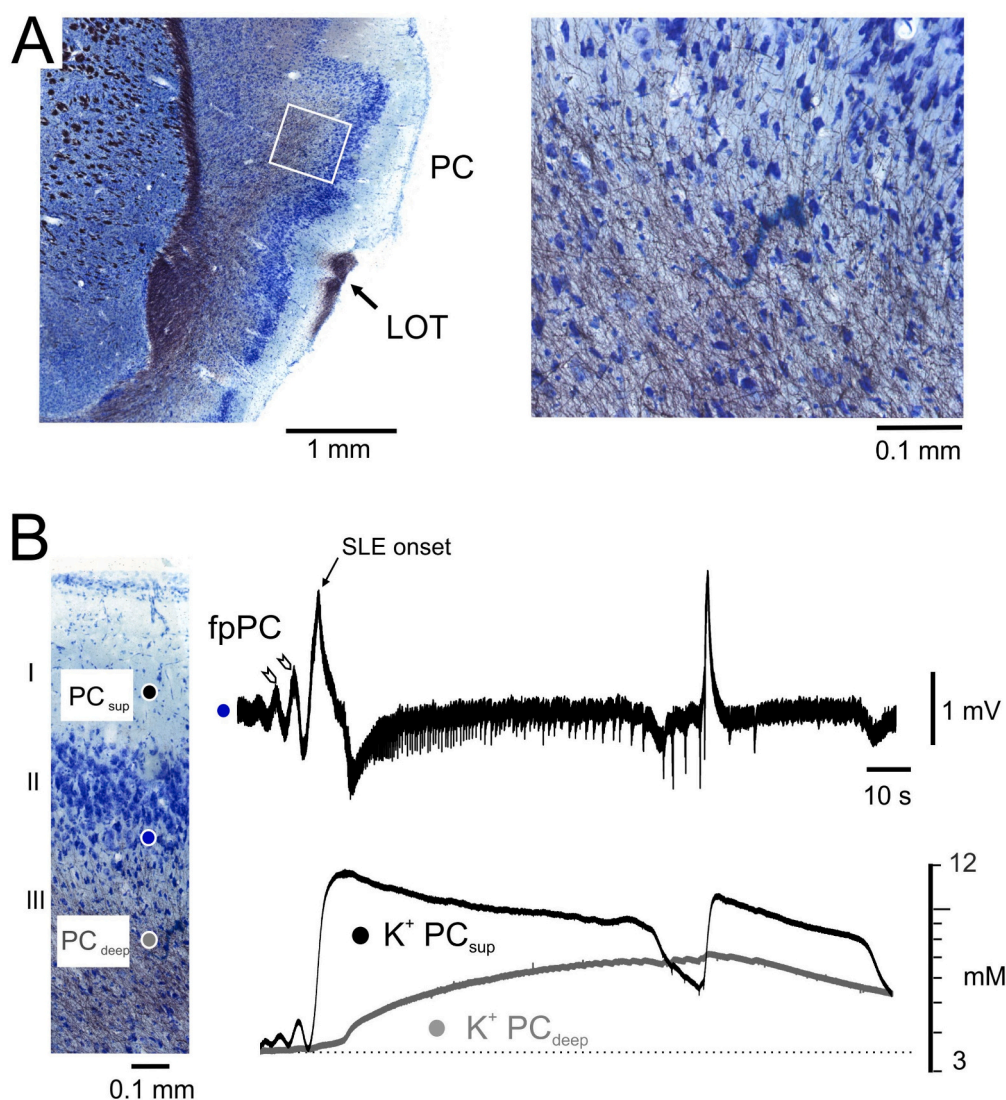


Fig. 1. Seizure-like events induced by 4-aminopyridine in the piriform cortex. **A.** Myelin-specific Black Gold II staining of a coronal section of a guinea-pig brain counterstained with thionin, showing the highly myelinated lateral olfactory tract (LOT) fibers and the piriform cortex (PC), where myelin staining is absent in superficial layer. The area outlined in the left photograph is expanded on the right, to illustrate the presence of black myelinated fibers in PC layers II and III. **B.** Typical seizure-like event (SLE) recorded in the PC during 50 μ M 4AP arterial perfusion. Extracellular field potential (upper trace fpPC, blue dot in the thionin/Black Gold II-stained histological picture on the left) and the simultaneously recorded potassium changes recorded with two K⁺-sensitive electrodes positioned in the unmyelinated superficial PC layer I (K⁺ PC_{sup}; black dot and lower black trace) and in the myelin enriched PC layer III (K⁺ PC_{deep}; grey dot and lower grey trace). The SLE onset (marked by the arrow) was typically preceded by slow oscillations (arrowheads; see also [Fig. 2A](#)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(adjusted with HCl). After 3-min perfusion the animal was decapitated, the brain was carefully dissected out (for details see de Curtis et al., 1991; Muhlethaler et al., 1993; de Curtis et al., 1998) and was placed in a recording chamber. A polyethylene cannula was inserted in the basilar artery, to restore the perfusion with the above saline solution (pH 7.4) at 5.5–6 ml/min. The brain is isolated with the resident pia intact and the solution perfused in the arterial system passively leaks from the main venous sinuses (see Librizzi et al., 1999). The temperature of the chamber was raised from 15 °C to 32 °C by 0.2 °C/min steps.

The experimental procedures and the animal care were conducted in accordance with the guidelines defined by the European Communities Council directive (2010/63/EU). Every effort was made to limit the number of animals used. The experimental protocol was reviewed and approved by the Ethics Committee and by the Committee on Animal Care and Use of the Fondazione IRCCS Istituto Neurologico Carlo Besta (Approval reference number: DO-01-2021).

2.2. Electrophysiological recordings and analysis

We utilized the PC as model region in which a highly myelinated structure, such as the lateral olfactory tract (LOT; arrow in the left panel in Fig. 1A), is in close proximity to the unmyelinated superficial fiber layer I (Fig. 1A and PC_{sup} in 1B) and the deep layers II–III that contain myelinated axons (Fig. 1A – right panel – and PC_{deep} in 1B). Field potential recordings from the *ex vivo* isolated brains were simultaneously performed in the LOT and the PC with glass capillaries filled with NaCl 0.9% (5–10 MΩ resistance). Responses evoked by stimulation of the LOT with a custom-made double-wire electrode connected to an isolation unit driven by a pulse generator (Telefactor S88; Grass, RI, USA) were monitored during the experiments. Trains of high frequency stimuli (hfST; internal frequency: 30 Hz; 1.5 s in duration, separated by intervals of 2–5 s) were delivered to test PC excitability. We utilized LOT stimulus intensities that generate half maximal disynaptic responses in the LOT-evoked PC response (between 8 and 9 Volts in different isolated brains) and then we increased it with steps of 1 V.

Electrophysiological signals were acquired at 3KHz sampling rate with a differential amplifier (Biomedical Engineering, Thornwood, NY, USA) and were digitized by National Instruments A/D board (AT-MIO-64E3, National Instruments, Italy). Electrophysiological signal acquisition and off-line analysis were performed with ELPHO® software (www.elpho.it) developed by Vadym Gnatkovsky in our lab.

To evaluate $[K^+]_o$ changes during neuronal activities, double barrel ion-sensitive electrodes were prepared. The capillary was exposed to dichlorodimethylsilane fumes (Fluka, Germany) and was heated for 1.5 h at 120 °C to fill the tip of one of the electrode barrel pair with a resin sensitive to K^+ (Potassium ionophore I, Cocktail A, Merck). The two barrels of the electrodes were then backfilled with either KCl 0.2 M (ion-sensitive electrode) or 0.9% NaCl (reference electrode barrel). Signals recorded from the reference and the ion sensitive electrodes were fed to a low-impedance differential amplifier (Supertech Instruments, Pecs, Hungary). The net $[K^+]_o$ increases were calculated by subtracting the reference from the ion sensitive signal. Before the experiment, the electrodes were calibrated using solutions with known concentration of K^+ . The measured voltage (in mV) was converted to K^+ concentration values using the formula $Y = s \log X$, where Y is the voltage measured, s is the slope of the calibration curve and X is the $[K^+]_o$ concentration (Librizzi et al., 2001). The plot of correlation between the K^+ changes and the speed of K^+ rise (first derivative) was computed with Origin Pro 2016 (OriginLab, Northampton, MA, USA).

To pool together different experiments for the analysis with approximately the same number of observations, for each experiment we selected one paired recording from each of the three following ranges of PC_{sup} 3.5–4 mM, 5–6.5 mM and 7–9 mM and the corresponding K^+ increases in both LOT or PC_{deep}. For the analysis of $[K^+]_o$ changes dynamics during hfST in PC_{sup}, LOT and PC_{deep} i) the mean K^+ shift values after each train were measured, and a curve fitting was performed

through the five mean values and ii) velocity- K^+ increases plots were performed using Origin Pro 2016 (OriginLab, Northampton, MA, USA).

2.3. Drugs

SLEs were induced by arterial application of the K^+ -channel blocker 4-aminopyridine (4AP; 50 μM; Tocris) for 5 min. Blockade of Kir4.1 K^+ channels was obtained by arterial perfusion of BaCl₂ (100 μM; Sigma-Aldrich) for 30 min.

2.4. Histological analysis

For histology, guinea-pig brains were fixed with 4% paraformaldehyde for 3–4 days. Coronal PC sections were cut by vibratome at 50 μm, were processed for myelin-specific Black Gold II staining (0.3% dissolved in 0.9% saline solution; Histo-Chem; Garbelli et al., 2012) based on the protocol provided on the company website. Sections were then counterstained with thionin (Sigma-Aldrich; Paletzki and Gerfen, 2019; Fig. 1A).

2.5. Statistical analysis

Statistical analysis was performed with Origin Pro 2016. Normal distribution was verified with Shapiro-Wilk test, if not differently specified, before application of the statistical tests. When the normality criterion was not satisfied non-parametric tests were chosen. Presence of outliers was checked with the Grubb's test. Significance was set at $p < 0.05$. Mean $\pm$ SD values are reported.

3. Results

$[K^+]_o$ rise is larger and faster in unmyelinated fibers layer than in the myelinated LOT during 4AP-induced SLEs. Arterial administration of 4AP (50 μM; 5 min) in the isolated *in vitro* -maintained guinea-pig brain induced a peculiar SLE pattern in the PC (Fig. 1B; see Carriero et al., 2010; Uva et al., 2017), preceded by 4–6 slow oscillations crowned by fast activity (arrowheads in Fig. 1B and 2A) that correlate with depolarization and discharge of layer II PC cells primarily (Uva et al., 2013). 4AP SLEs in the PC are accompanied by fast $[K^+]_o$ increases in superficial layer I (K^+ PC_{sup} in Fig. 1B), which is composed exclusively of both unmyelinated fibers that leave the highly myelinated LOT (arrow in Fig. 1A) and unmyelinated intra-PC associative fibers (for review see Neville and Haberly, 2004). $[K^+]_o$ changes in PC_{sup} were larger and faster than in deeper cellular layers II–III (K^+ PC_{deep} in Fig. 1B) that contain densely arranged neurons intermingled with myelinated axons, as illustrated in Fig. 1A (right panel) and 1B.

In a first set of experiments, 16 isolated guinea-pig brains were perfused with 4AP and simultaneous measurements of $[K^+]_o$ in the unmyelinated PC_{sup} (layer I) and in the myelinated LOT during the SLEs were performed, as illustrated in Fig. 2A. We observed that the $[K^+]_o$ in PC_{sup} started to increase either together or before LOT ($\Delta t = 3.2 \pm 4.5$ s; Fig. 2B). The $[K^+]_o$ shifts were consistently and significantly higher in PC_{sup} (black trace and dot in the left insert of Fig. 2A) compared to LOT (grey trace and dot in the left insert), both at SLE onset ($[K^+]_o$ in PC_{sup} = 3.86 ± 0.58 mM; $[K^+]_o$ in LOT = 3.49 ± 0.35 mM; Wilcoxon Signed Rank test, $Z = 2.197$, $p = 0.0245$; Figure 2Ca) and at the K^+ peak ($[K^+]_o$ in PC_{sup} = 6.67 ± 2.45 mM; $[K^+]_o$ in LOT = 4.91 ± 1.28 mM; Wilcoxon Signed Rank test, $Z = 2.922$, $p = 0.00168$; Figure 2Cb). These experiments demonstrated that the accumulation of $[K^+]_o$ is larger and faster in unmyelinated regions during 4AP-induced SLEs.

$[K^+]_o$ increase is faster in the unmyelinated fibers PC_{sup} during high frequency electrical stimulations. In the next set of experiments, we mimicked by LOT stimulation the preictal phase that precedes the typical PC SLE pattern induced by arterial application of 4AP (illustrated in Figs. 1B and 2A). In 10 experiments we delivered 5 trains of high frequency electrical stimuli (hfST) to the LOT every 2–5 s with an

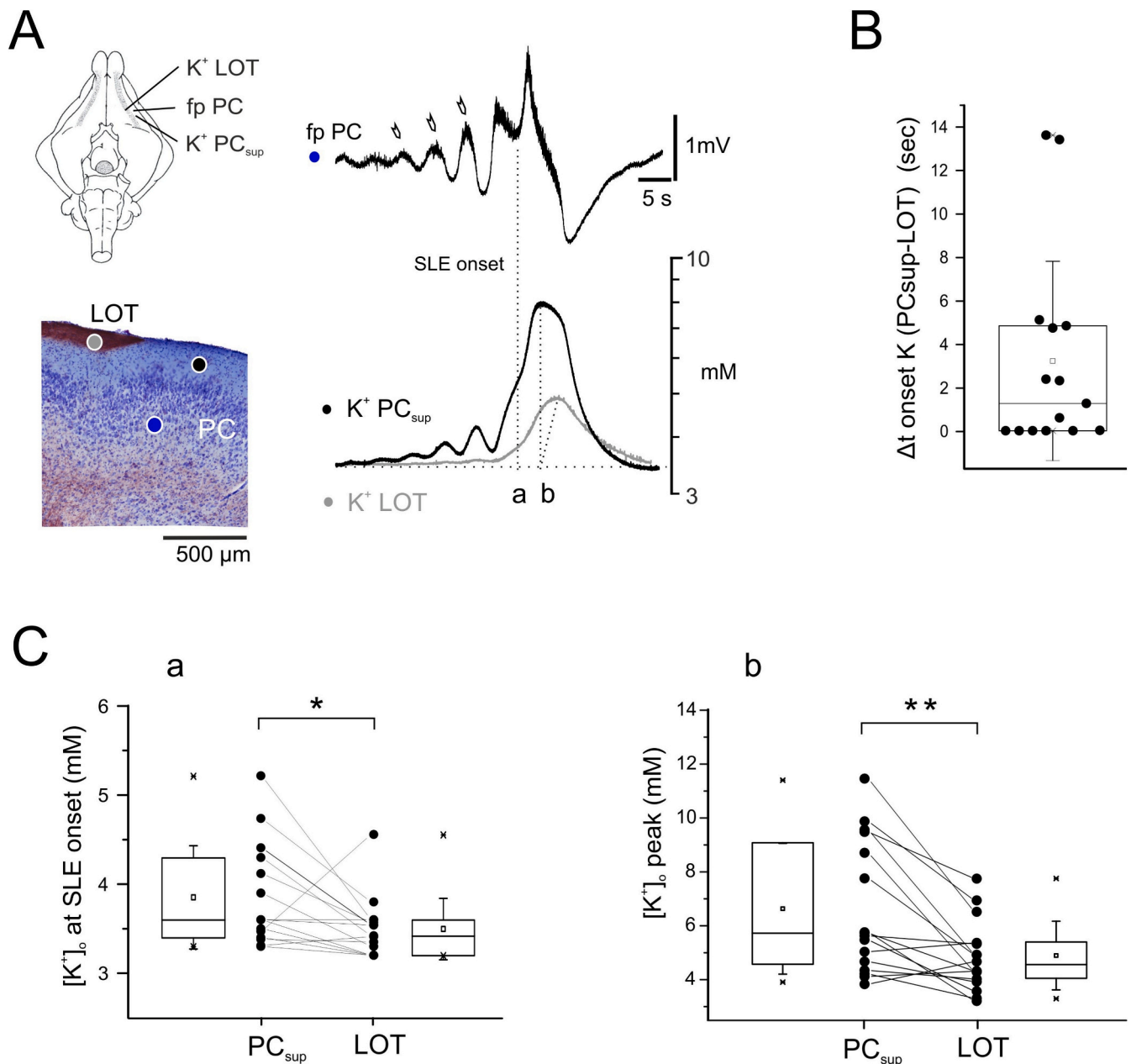


Fig. 2. Larger $[K^+]_o$ changes in superficial PC layer during seizure-like events (SLEs) induced by 4AP in the PC. **A.** Scheme of a ventral view of a guinea pig brain with the position of the field potential (fp) recording electrode (blue dot) and the K^+ sensitive electrodes in the layer I of the PC (PC_{sup}) and the LOT – black and grey dots, respectively, on a coronal section of the PC stained for thionin and Black Gold II. Simultaneous recording of a representative SLE (upper trace) and the relative changes in $[K^+]_o$ in PC_{sup} (black trace) and in the LOT (grey trace). Typical pre-SLEs slow oscillations (arrowheads) are shown. Vertical dotted lines mark the SLE onset (a) and the peak $[K^+]_o$ deflection (b). **B.** Quantification of the rise time delay in LOT and PC_{sup}, measured as the difference between $[K^+]_o$ shift onsets. **C.** $[K^+]_o$ at SLE onset (**Ca**) and at $[K^+]_o$ peak (**Cb**). The pairs of measurements simultaneously performed in the same experiments ($n = 16$) are illustrated in middle part of the graphs in **Ca** and **Cb**. Statistical significance is reported. Wilcoxon Signed Rank test, $p = 0.0245$ at onset and $p = 0.00168$ at the K^+ peak. *, $p < 0.05$; **, $p < 0.01$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

internal frequency of 30 Hz and train duration of 1.5 s (upper left scheme in Fig. 3A). Evoked responses were recorded in PC and $[K^+]_o$ levels were simultaneously monitored in the unmyelinated layer I (PC_{sup}) and in the highly myelinated LOT (black and grey lower traces, respectively, in Fig. 3A). The $[K^+]_o$ increases in both PC_{sup} and LOT were proportional to the voltage intensity of the LOT hfSTs ($n = 5$ brains), as quantified in Fig. 3B; in two experiments, high amplitude hfSTs resulted in epileptiform afterdischarges that outlasted the hfST duration (open arrow in the right panel in Fig. 3A) and were associated to a prolongation of $[K^+]_o$ increase (black arrow in the lower black trace). hfSTs inducing afterdischarges were excluded from the analysis shown in

Fig. 3.

Following hfST protocol, $[K^+]_o$ shifts were statistically larger in PC_{sup} compared to LOT ($n = 19$ in 10 brains; PC $[K^+]_o$ peak: 5.76 ± 1.73 mM; LOT $[K^+]_o$ peak: 4.09 ± 0.77 mM; Fig. 3C; Wilcoxon Signed Ranks Test, $Z = -3.321$, $p = 0.000248$) regardless of the intensity of the stimuli and the interval between trains.

Next, we analyzed the dynamic of the $[K^+]_o$ modifications in the pairs of PC_{sup}-LOT recordings in which hfST-induced PC_{sup} $[K^+]_o$ was >5 mM ($n = 11$ in 7 brains). After each train within the hfST, $[K^+]_o$ profile recorded in the PC_{sup} displayed an increase and a decrease that did not return to background values (Fig. 4). This pattern is better

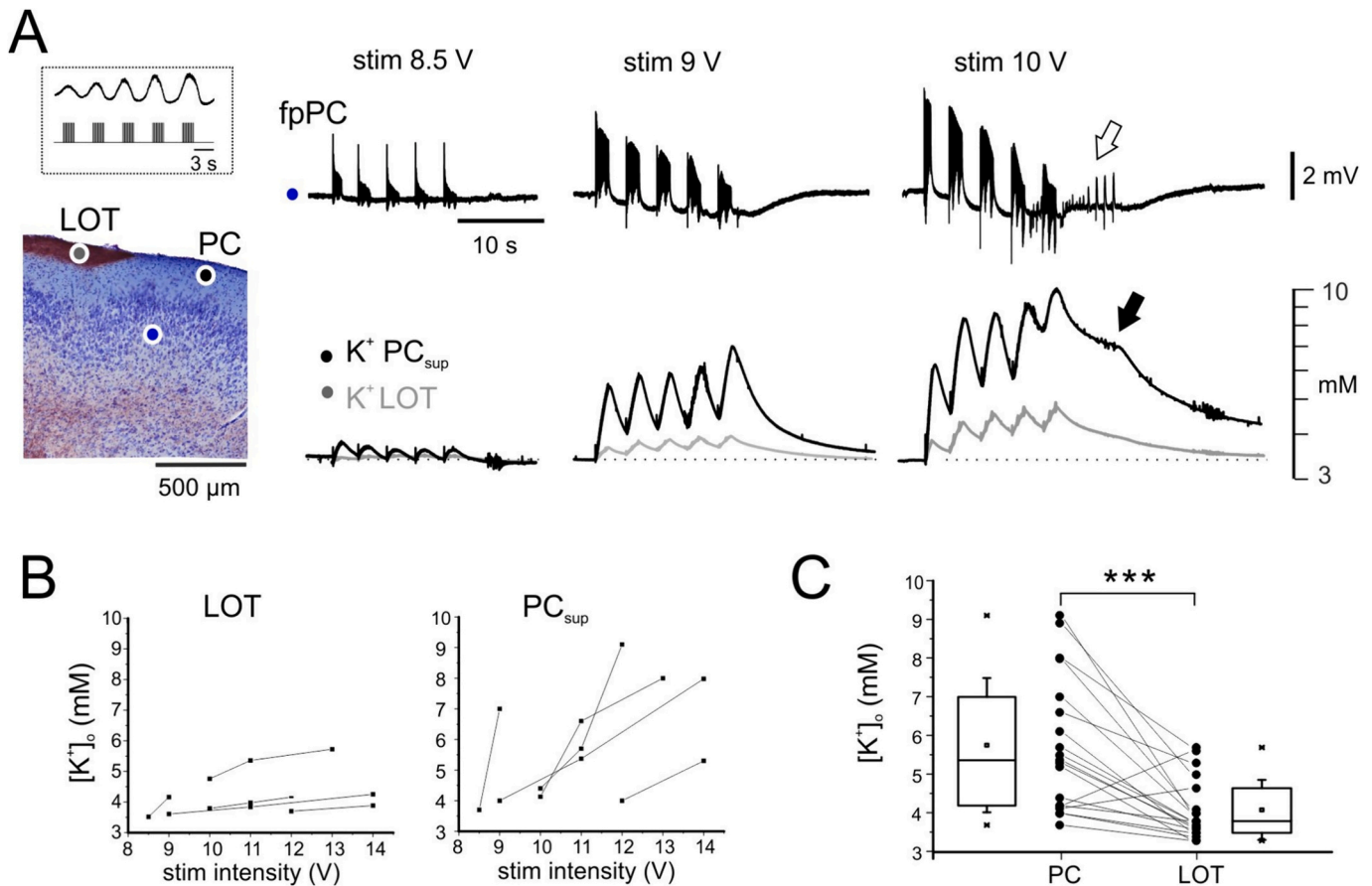


Fig. 3. Extracellular potassium changes during repetitive high-frequency LOT stimuli. **A.** On the left, a scheme of the LOT stimulation that mimics the 4 AP-induced pre-SLE activation (represented in the upper trace): 5 trains of 30 Hz electrical stimuli delivered for 1.5 s, every 3.5 s (lower trace). The position of the simultaneous K^+ recording in PC_{sup} (black dot) and in LOT (grey dot) and field potential recording electrode (blue dot) are shown in the micrograph. Field potentials (upper traces) and $[K^+]_o$ changes recorded in PC_{sup} (black trace) and LOT (grey trace) at three different stimulation intensities of 8.5 V (left traces), 9 V (middle traces) and 10 V (right traces). The open arrow points to a hfST-induced afterdischarge and the black arrow to the associated hump in $[K^+]_o$ rise. **B.** $[K^+]_o$ changes simultaneously recorded in LOT (left) and PC_{sup} (right) at the 2–3 stimulus intensities. **C.** $[K^+]_o$ rises in LOT (right plot) and PC_{sup} (left plot) during hfST. In the middle part of the panel, lines connect the K^+ peaks of pairs of simultaneously K^+ measurement in the two regions. Maximal LOT peaks are significantly lower than in PC_{sup} ($n = 19$ in 10 brains; Wilcoxon Signed Ranks Test, $p = 0.000248$). ***, $p < 0.001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

illustrated by the analysis described in the right panel of Fig. 4A, where the velocity of the $[K^+]_o$ changes was plotted as a function of the $[K^+]_o$; the K^+ increment at each of the 5 single hfST train were followed by a partial return toward the baseline. Interestingly, the highest $[K^+]_o$ shifts and velocities in PC_{sup} were associated to the first and the second train ($n = 9$ out of 11 train stimulations) indicating a rapid $[K^+]_o$ rise in the tissue containing unmyelinated fibers. In 5 experiments, $[K^+]_o$ did not further increase after the 2nd or 3rd train (as illustrated by the black traces and lines in the typical PC_{sup} examples in Fig. 4A and Fig. 5A), suggesting a plateau value of K^+ rise, likely due to local buffering mechanisms. In the LOT, $[K^+]_o$ modifications showed either the same incremental pattern described in the PC_{sup}, with smaller velocity and $[K^+]_o$ values ($n = 3$), or a progressive K^+ accumulation with each subsequent hfST train ($n = 8$; grey traces and line in Fig. 4A).

In order to quantify the trend described above, peak $[K^+]_o$ amplitude was measured during the 5 consecutive train bursts that form the hfST. We found that average $[K^+]_o$ peaks increased during subsequent hfST trains following curves described by polynomial equations:

$y = -0.0523 \times x^2 + 0.6978 \times x + 4.422$, $R^2 = 0.9943$ in PC_{sup} and $y = -0.02 \times x^2 + 0.2615 \times x + 3.336$, $R^2 = 0.9974$ in LOT. These findings are reported in Fig. 4B.

In another series of experiments, we compared the $[K^+]_o$ changes during LOT hfST at two different depths in layer I (PC_{sup}) versus layer II-

III at 500–750 μ m depth (PC_{deep} or PC₅₀₀ and PC₇₅₀; $n = 13$ brains; Fig. 5). The peak of $[K^+]_o$ associated to electrical stimulation proportionally increased with the stimulus intensity (Fig. 5B) and were significantly lower in PC_{deep} (3.84 ± 0.77 mM in PC₅₀₀, $n = 21$ hfSTs; 3.86 ± 0.35 mM in PC₇₅₀, $n = 8$ hfSTs) compared to the surface PC_{sup} (5.63 ± 1.60 mM, $n = 28$ hfSTs). Statistical analysis of the simultaneously recorded pairs demonstrated a significant difference of PC_{sup} vs PC₅₀₀ (Wilcoxon Signed Ranks Test, $Z = 3.997$, $p < 0.0001$, Fig. 5C) and PC_{sup} vs PC₇₅₀ (Wilcoxon Signed Ranks Test, $Z = 2.450$, $p = 0.00781$; Fig. 5C). No statistical difference in $[K^+]_o$ increments was found between PC₅₀₀ and PC₇₅₀ (Mann-Whitney Test, $U = 62.5$, $p = 0.304$; Fig. 5C). For these quantifications, hfST that induced epileptiform afterdischarges were excluded.

The different $[K^+]_o$ pattern between PC_{sup} and PC_{deep} during hfST was further highlighted by the analysis of the rising velocity of $[K^+]_o$ changes as a function of $[K^+]_o$ during the 5 subsequent trains (lower panel in Fig. 5A; $n = 14$ tests in 9 brains). The plots obtained in PC_{sup} showed a faster $[K^+]_o$ increase compared to PC_{deep} ($n = 7$). As represented in the lower panel of Fig. 5A PC_{deep} (grey lines) showed a slower and progressive $[K^+]_o$ increase during the train stimulations ($n = 13$).

Quantification of the $[K^+]_o$ peaks after each subsequent hfST in the simultaneous PC_{sup} and PC_{deep} recordings ($n = 17$ in 11 brains) confirmed larger $[K^+]_o$ changes in the PC_{sup} compared to deep PC (PC₅₀₀

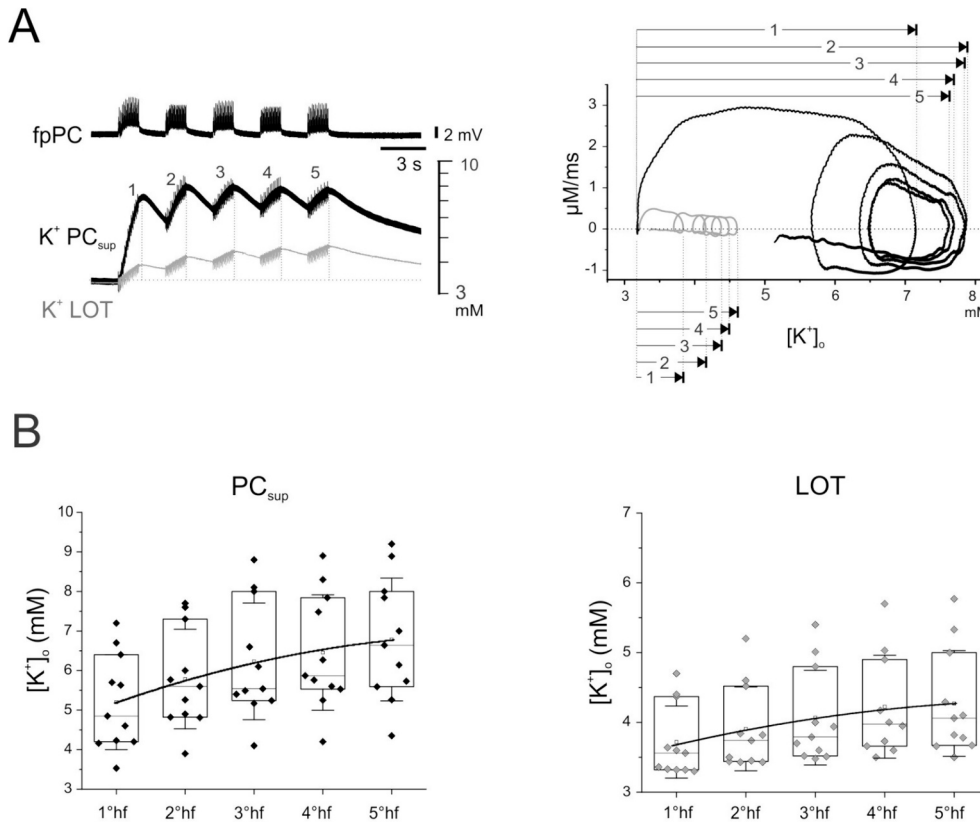


Fig. 4. Extracellular potassium changes in PC_{sup} and LOT during the repetition of five high-frequency LOT stimuli that mimic the pre-SLE activity. **A.** Field response (fpPC upper trace) and [K⁺]_o changes induced by the 5 consecutive trains during a hfST (lower traces). In the right panel, the velocity of [K⁺]_o changes are plotted against the [K⁺]_o values for both positions (PC_{sup} in black and LOT in grey). Each loop represents one single train of the 5 of a hfST cycle, from the first (1) to the last (5). **B.** Averaged peak extracellular K⁺ changes during the 5 subsequent bursts forming the hfST protocol ($n = 11$ in 7 brains) in the PC_{sup} (left) and in the LOT (right). Polynomial equations describe the average increase of during hfST: $y = -0.0523 \times x^2 + 0.6978x + 4.422$, $R^2 = 0.9943$ in PC_{sup} and $y = -0.02 \times x^2 + 0.2615x + 3.336$, $R^2 = 0.9974$ in LOT.

and PC₇₅₀ in Fig. 5C). Interestingly, the K⁺ rise in the PC_{sup} fitted with a polynomial curve ($y = -0.0835 \times x^2 + 1.0195x + 3.5955$, $R^2 = 0.999$), whereas in PC_{deep} the increases in [K⁺]_o were described by a linear function ($y = 0.152x + 3.438$, $R^2 = 0.9974$).

The [K⁺]_o decay is faster in superficial PC than in deep PC. To compare the decay of [K⁺]_o in surface and deep PC positions, we measured the K⁺ levels at the end and 7 s after the hfST (vertical dotted lines in Fig. 6A). The ratio between the two measurements indicates a significantly faster K⁺ decay in PC_{sup} than in PC_{deep} ($36.63 \pm 18.55\%$ and $58.98 \pm 26.80\%$ of the [K⁺]_o measured at the end of the hfST, respectively; paired sample t -test, $t(18) = -3.42$, $p = 0.00302$; on the left in Fig. 6B), suggesting a faster [K⁺]_o buffering in PC_{sup} compared to PC_{deep}. In 7 hfST from 5 experiments, trains of stimuli resulted in post hfSTs afterdischarge that outlasted hfST (arrows in Fig. 6A); in these experiments the PC_{sup} [K⁺]_o partially recovered 7 s after the end of the hfST ($59.51 \pm 27.61\%$), whereas in PC_{deep} [K⁺]_o further increased during the afterdischarge ($108.72 \pm 39.97\%$); the difference between the K⁺ decays computed in PC_{sup} and PC_{deep} was statistically different (paired sample t -test, $t(6) = -2.925$, $p = 0.0265$; Fig. 6B). These data show that the hyperexcitability associated to an afterdischarge induced a prolongation of [K⁺]_o increase and a further augmentation mainly in the deep PC sites.

The blockade of the Kir4.1 channels increases [K⁺]_o shifts in both superficial and deep PC layers and decreases the excitability threshold. Kir4.1 K⁺ channels are present on both astrocytes membrane and oligodendrocyte myelin inner tongue (Schirmer et al., 2018) to remove from the extracellular space the K⁺ released during neuronal activity. The blockade of Kir4.1 channels is expected to decrease the efficacy of K⁺ buffering. We perfused the Kir4.1 K⁺ channel blocker BaCl₂ (100 μM

for 30 min) in 7 isolated brains, and in 5 we simultaneously recorded the hfST-evoked [K⁺]_o shifts in PC_{sup} and PC_{deep}. LOT hfST were delivered before and at the end of the perfusion with BaCl₂ ($n = 11$) at the same amplitude to evaluate [K⁺]_o changes induced in both PC_{sup} and PC_{deep} (Fig. 7). A larger [K⁺]_o increase was observed in the two PC positions after the BaCl₂ administration; maximal [K⁺]_o in PC_{sup} were 6.22 ± 1.64 mM and 8.48 ± 1.76 mM in control conditions and after BaCl₂ perfusion, respectively (paired sample t -test, $t(10) = -6.3024$, $p < 0.0001$). In PC_{deep} control [K⁺]_o rise reached 3.98 ± 0.97 mM and after BaCl₂ perfusion increased to 5.90 ± 2.47 mM (Wilcoxon signed rank test, $Z = -2.891$, $p = 9.766 \text{ E-}4$). The [K⁺]_o shifts recorded in PC_{sup} and PC_{deep} before and after BaCl₂ perfusion were statistically different (Wilcoxon Signed Ranks Test, $Z = 2.889$, $p = 9.766 \text{ E-}4$ in control condition and $Z = 2.650$, $p = 0.00391$ after BaCl₂ perfusion) and the mean percentage of [K⁺]_o increase measured during BaCl₂ perfusion at the peak in the two sites increased by 37% and 54%, respectively (Fig. 7C). The application of BaCl₂ also decreased the amplitude of train stimuli necessary to induce afterdischarges. As illustrated in the right panels in Fig. 7A and B, the presence of BaCl₂ increased the [K⁺]_o rise observed after each train and induced [K⁺]_o accumulation during the hfST progression in both PC_{sup} (black traces) and PC_{deep} (grey traces).

The [K⁺]_o at the end of the last hfST and 7 s after was measured in both PC_{sup} and PC_{deep} control condition and after BaCl₂ application (Fig. 7D, n brains = 5). The return to baseline appeared slower in both PC_{sup} and PC_{deep} after BaCl₂ administration. Two-way ANOVA test for repeated measures was utilized, after confirmation of normal distribution of the populations with Kosmogorov-Smirnov test, that revealed a significant effect of the two factors “site of recording” ($F(1) = 9.851$, $p = 0.00331$) and “treatment with BaCl₂” ($F(1) = 18.160$, $p = 0.01137$) on

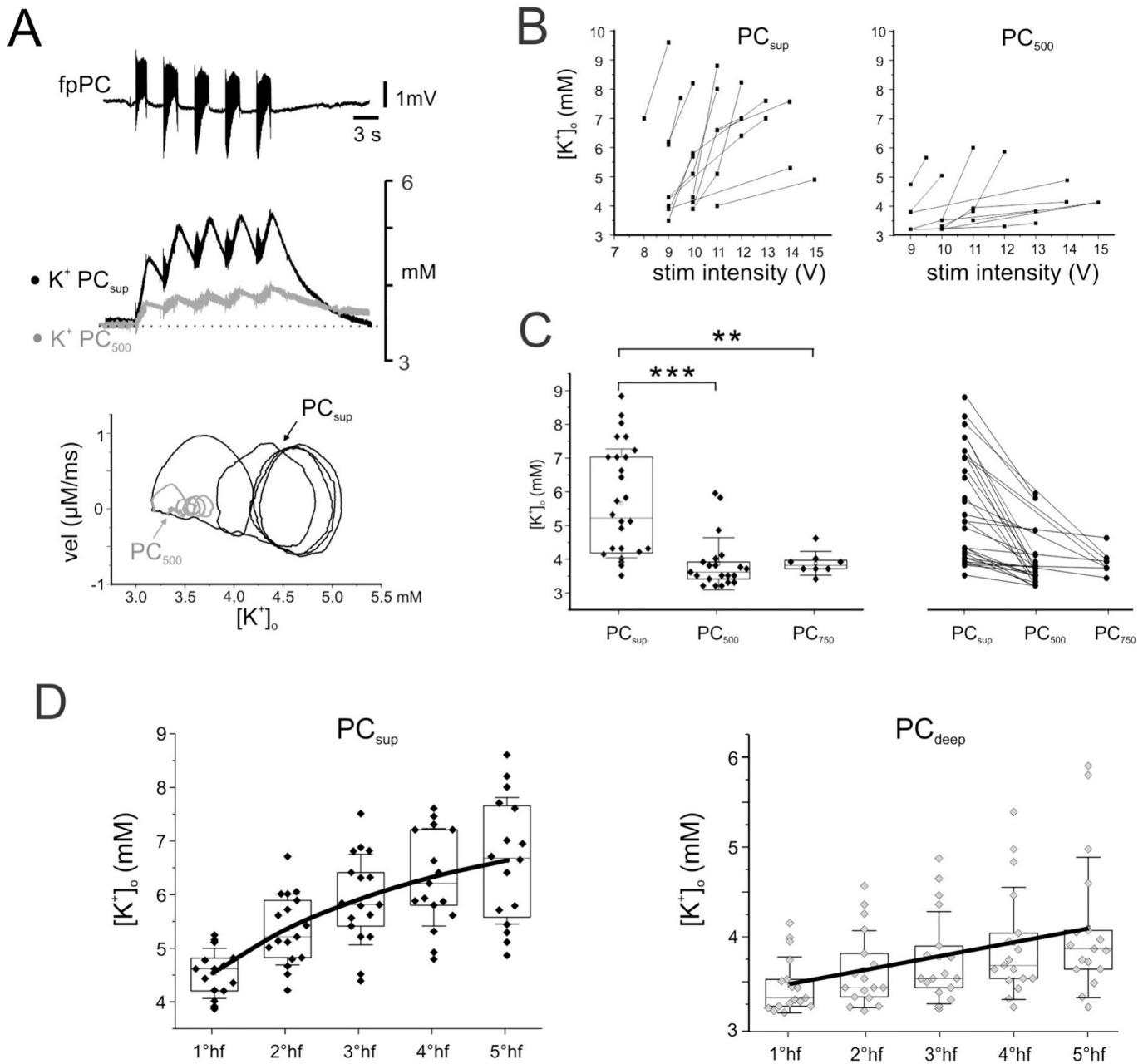


Fig. 5. Extracellular potassium changes in PC_{sup} and PC_{deep} during high-frequency LOT stimulation. **A.** $[K^+]_o$ changes in PC_{sup} and PC_{deep} recorded at 500 μm (PC₅₀₀) and 750 μm (PC₇₅₀) from the pial surface. Responses to 5 consecutive high-frequency LOT stimuli that mimic the pre-SLE oscillations (see Fig. 3) are shown. At the bottom, the K^+ concentration/velocity plot recorded in PC_{sup} (black line) and PC₅₀₀ (grey line). **B.** Correlation between $[K^+]_o$ changes in PC_{sup} and PC₅₀₀ and stimulus intensities. Connecting lines identify values recorded in the same experiments at two or three different hfST intensities. **C.** Absolute values of $[K^+]_o$ changes in PC_{sup}, PC₅₀₀ and PC₇₅₀ (left panel); on the right, the values simultaneously recorded in the same position at two or three sites are connected by a line. PC_{sup} vs PC₅₀₀: Wilcoxon Signed Ranks Test, $p < 0.0001$; PC_{sup} vs PC₇₅₀: Wilcoxon Signed Ranks Test, $p = 0.00781$; PC₅₀₀ and PC₇₅₀: Mann-Whitney Test, $p = 0.304$. **D.** Progressive $[K^+]_o$ increases during hfST recorded in PC_{sup} and PC_{deep} ($n = 17$ in 11 brains) are described by a polynomial curve ($y = -0.0835 \times x^2 + 1.0195 \times x + 3.5955$, $R^2 = 0.999$; left panel) and by a linear function ($y = 0.152 \times x + 3.438$, $R^2 = 0.9974$; right panel), respectively. **, $p < 0.01$; ***, $p < 0.001$.

$[K^+]_o$ decays; interaction between site of recording and treatment with BaCl₂ was evaluated ($F(1) = 0.0305$) and Tukey *post hoc* test showed significant pairwise differences in PC_{deep} in control condition vs treatment with BaCl₂ ($p = 0.0417$), between PC_{sup} and PC_{deep} in control condition ($p = 0.0199$) and after BaCl₂ treatment ($p = 0.0159$), whereas in PC_{sup} a trend close to significance was found ($p = 0.0523$). The blockade of K^+ buffering operated by BaCl₂ and the associated hyperexcitability determined longer times to baseline in PC_{deep} compared to PC_{sup} (Fig. 7D), as observed during the afterdischarges induced by hfST in control conditions.

4. Discussion

The three-layered PC has a peculiar intrinsic organization extensively analyzed in different mammals (Haberly, 2001; Neville and Haberly, 2004; de Curtis et al., 2019, 2023). The PC receives a direct input from axons originating from olfactory bulb neurons that form the fully myelinated fiber bundle of the LOT; entering the PC, these afferent axons lose their myelin coating and terminate on the apical dendrites of layer II-III neurons in the most superficial layer I (layer Ia). In the deeper part of the superficial layer I (named Ib) runs another laminar band of

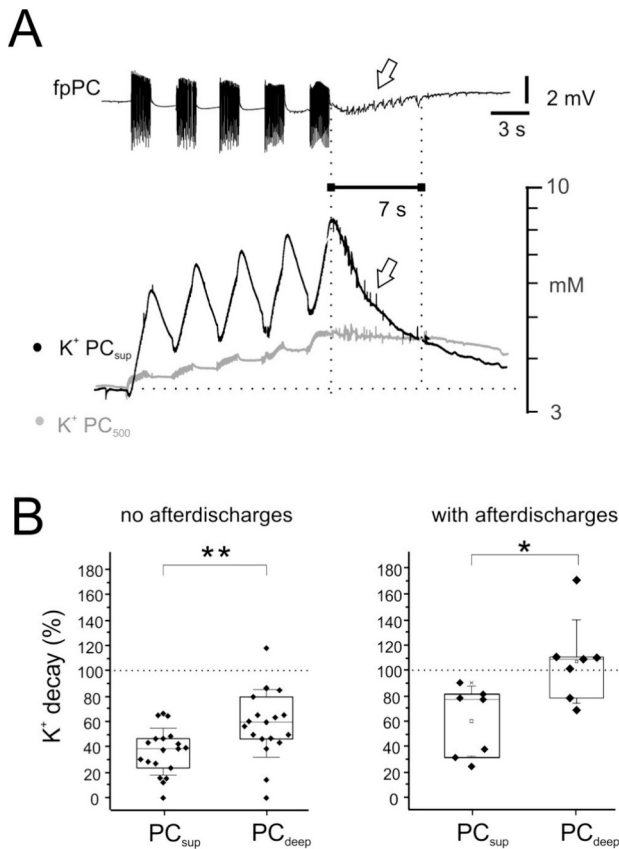


Fig. 6. Recovery of extracellular potassium induced in different PC layers by high-frequency LOT stimulation. A. Field response (upper trace) and $[K^+]_o$ changes in PC_{sup} and PC_{500} (black and grey lower traces, respectively) during 5 cycles of hfST, as illustrated in Fig. 5A. In the illustrated experiment, hfST induced an afterdischarge coupled with a $[K^+]_o$ increase (arrows). Vertical dotted lines represent time points after hfST-separated by 7 s – when $[K^+]_o$ measurements were performed. B. $[K^+]_o$ decay changes measured in PC_{sup} and PC_{deep} 7 s after the end of the hfST (dotted lines in panel A) in experiments without (left panel; $n = 19$ from 10 brains; paired sample t -test, $p = 0.00302$) and with afterdischarges after hfST (right panel; $n = 7$ from 5 brains; paired sample t -test, $p = 0.0265$). *, $p < 0.05$; **, $p < 0.01$.

unmyelinated fibers formed by the intra-PC associative axons connecting layer II-III neurons within the PC. Notably, layer I contains very sparse interneurons (Suzuki and Bekkers, 2010). Local connections within layers II and III neurons are supported by myelinated axons. The unique spatial segregation of axons in the PC and LOT elects this as the ideal cortical region to study K^+ dynamic in organized bundles of unmyelinated (layer I) and myelinated (LOT and deep layers) fibers. The neurophysiological networks of the PC and other olfactory regions has been extensively analyzed in the *ex vivo* brain preparation we utilized for the present study (Biella and de Curtis, 1995, 2000; Uva and De Curtis, 2005; Carriero et al., 2009).

We simultaneously recorded extracellular activities at three close sites i) in the myelinated LOT, ii) in the virtually acellular PC layer I formed by densely arranged unmyelinated fibers and iii) in PC deep layers II and III that contain neurons and myelinated fibers. LOT input is homogeneously distributed in the PC, which is therefore simultaneously activated when LOT is stimulated. Associative corticocortical intrinsic PC connections are also very broadly distributed within PC. Synaptic connections and densities in layers Ia and Ib and in deeper layers are homogeneously distributed in the PC and no variations are expected within the PC in the rostro-caudal direction.

Our experiments demonstrate that a) $[K^+]_o$ changes induced by 4AP SLEs are larger in unmyelinated than in myelinated areas, b) patterned

high-frequency stimulation trains of the LOT that mimic pre-SLE activity generate larger $[K^+]_o$ rises in the unmyelinated superficial layer I in comparison to the myelinated LOT and the deep PC layers, c) $[K^+]_o$ decay is slower in deep than in superficial PC layers; d) the blockade of Kir4.1 channels by $BaCl_2$ enhances hfST-induced $[K^+]_o$ rise in both superficial and deep layers; e) the latter effect is more evident in deep myelinated PC layers. Incidentally, we show that $[K^+]_o$ increases in deep PC layers (but not in superficial layer I) when afterdischarges are generated by both hfST and SLEs in layer II and III neurons (see also Uva et al., 2017).

In myelinated fibers, the inwardly rectifying K^+ (Kir) 4.1 channels located in the inner tongue of the myelin sheath facing the axon membrane beneath juxtaparanodal myelin (Wang et al., 1993; Rash, 2010; Schirmer et al., 2018; Giaume et al., 2021) removes the excess of extracellular K^+ released during hypersynchronous neuronal firing (David et al., 1993; Poopalasundaram et al., 2000; Larson et al., 2018). The oligodendrocytes that form the myelin either block the diffusion of K^+ in the extracellular space or are interconnected with other oligodendrocytes and with astrocytes via gap junctions, to form a panglial syncytium (Rash, 2010; Tress et al., 2012). The extracellular K^+ taken up by the oligodendrocytes is eliminated through the gap junction-interconnected panglial network (Kettenmann and Ransom, 1988; Butt and Ransom, 1989), and in particular by astrocyte endfeet via Kir4.1 channels (Newman, 1986; Chaban et al., 2017; Beckner, 2020; Giaume et al., 2021). We can assume that K^+ increases during brain activity between the axonal membrane and myelin are higher than the variations that we recorded with our extracellular electrodes, but we have not direct access to that space. No less important, myelin induces a segregation of the ion fluxes associated to the propagation of the action potentials along axons and its repolarization, that are instead uniformly distributed in unmyelinated axons.

We mimicked the periodic preictal events with repeated high frequency trains. The induced activation is not identical to the preictal oscillations induced by 4AP application, because it determines a synchronized activation of the LOT fibers that is not likely to occur during 4AP, but we confirmed a smaller K^+ increase in both LOT and PC_{deep} compared to PC_{sup} unmyelinated layer I. In 4AP-induced SLEs, we interpreted the lower and slower $[K^+]_o$ observed in PC_{deep} as due to the diffusion of the K^+ released in the PC_{sup} (Uva et al., 2017); still, the rise of $[K^+]_o$ during LOT hfST could also be due to the direct activation of layer II-III neurons.

The $[K^+]_o$ increase in layer I measured at the onset of hfST rapidly reached a plateau concentration within the first 2–3 trains of stimuli; interestingly, this dynamic resembles the fast increase observed at SLE onset. The onset of both SLE and hfST $[K^+]_o$ shifts were smaller and slower in PC_{deep} , possibly because of a more controlled and gradual K^+ release in the extracellular space due to the different organization of glia and the axonal myelination in the different PC layers.

Martin-Lopez et al. (2019) showed that the number of the astrocytes in the mice PC is different across layers, with the highest density in layer Ia and the lowest in layer II. Moreover, in mice, a combination of $\alpha 2$ and $\beta 2$ subunits of the Na^+-K^+ ATPase are expressed by astrocytes (Murata et al., 2020). This isoforms combination controls the K^+ homeostasis during neuronal activity (Larsen et al., 2016) to promptly buffer the rise in K^+ during hfST and SLE observed in our experimental conditions; this could explain the rapid return of K^+ to the baseline after each train in layer I. The above mentioned lower density of astrocytes in PC_{deep} compared to layer I (Martin-Lopez et al., 2019) could be responsible of the progressive and continuous rise of $[K^+]_o$ in PC_{deep} during afterdischarges (as discussed below). The lower number of astrocytes could also be responsible for the slower $[K^+]_o$ decay in PC_{deep} compared to the PC_{sup} observed in our experiments.

Afterdischarges could be generated as late events during both SLEs and hfSTs. The spikes recorded during such afterdischarges were generated in the depth of the cortex, where neurons of layer II and III are located (Uva et al., 2013, 2017). Such synchronized neuronal activity

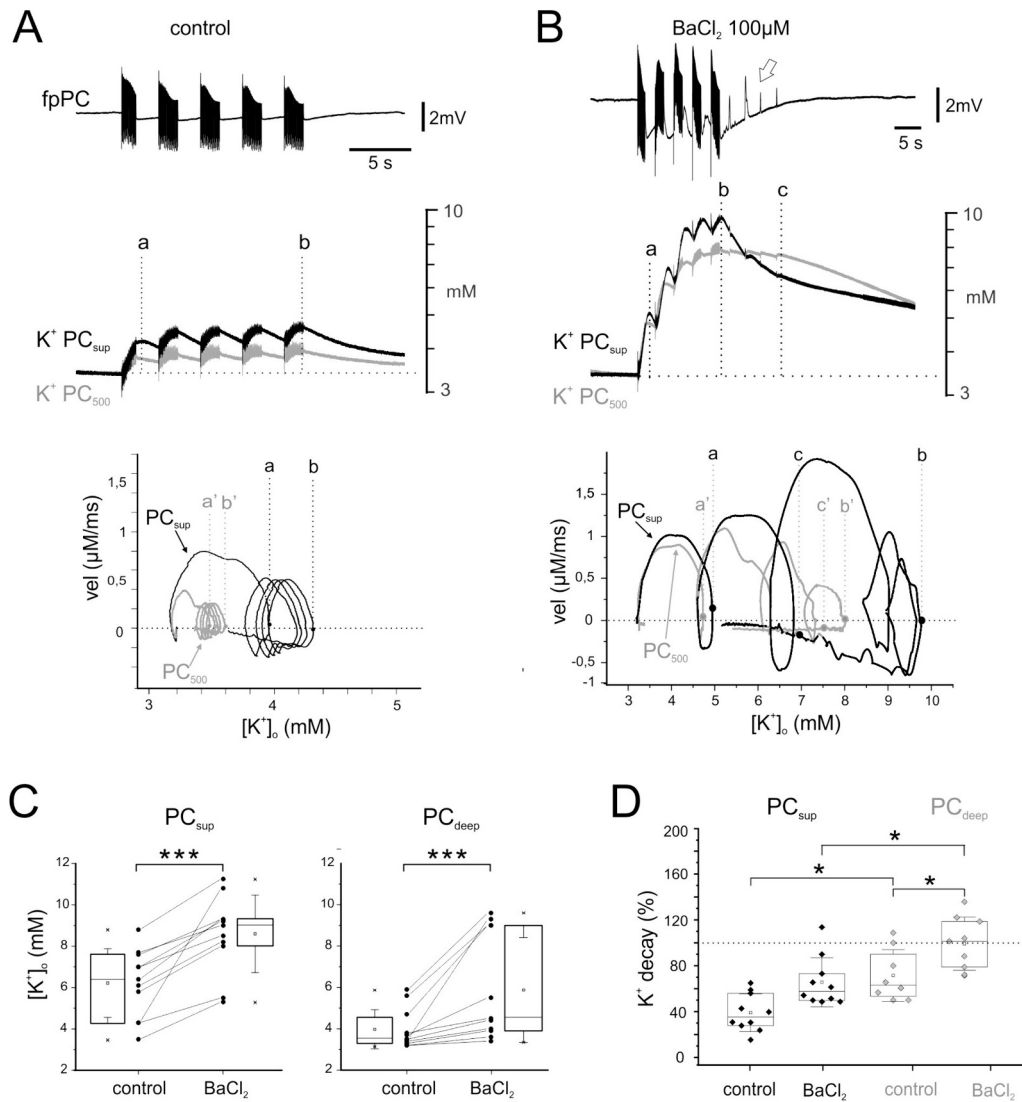


Fig. 7. Effect of the Kir4.1 channel blocker BaCl₂ on hfST-induced $[K^+]_o$ changes in different PC layers. **A.** Field response (upper trace) and $[K^+]_o$ measurements in the PC_{sup} (black trace) and PC₅₀₀ (grey trace) during LOT hfST. On the bottom panel, $[K^+]_o$ increases are plotted against its rate of change, as illustrated in Fig. 5. **B.** LOT hfST amplitude and recording sites are the same as in A, after brain perfusion with 100 μ M BaCl₂. **C.** $[K^+]_o$ values in PC_{sup} (left) and PC_{deep} (right) before and after BaCl₂ treatment show a significant increase of $[K^+]_o$ after blockade of Kir4.1 channels ($n = 11$ from 7 brains; PC_{sup}: paired sample t-test, $p < 0.0001$; PC_{deep}: Wilcoxon signed rank test, $p = 9.776 \times 10^{-4}$). $[K^+]_o$ values before and after BaCl₂ in the same experiments are illustrated by connected lines. **D.** $[K^+]_o$ decay changes measured in PC_{sup} and PC_{deep} 7 s after the end of the hfST before and after BaCl₂ treatments, after 10 hfST in 5 brains. In both regions a slower decay was observed after BaCl₂ perfusion (two-way ANOVA for repeated measures followed by *post hoc* Tukey test). *, $p < 0.05$.

was associated to a further K^+ increase in PC_{deep}, while $[K^+]_o$ returned to baseline in PC_{sup}, where no discharging neurons are found. We propose that the different $[K^+]_o$ dynamics between PC_{sup} and PC_{deep} sites during both the afterdischarge and the late SLE could be explained by the different anatomy and by the location of the generators of neuronal spiking activity.

As in Bay and Butt (2012), the pharmacological manipulation with the Kir4.1 channel blocker BaCl₂ induced 37% and 54% increase of maximal $[K^+]_o$ rise in both superficial and deep layers, respectively, and decreased the threshold for the generation of afterdischarges at the end of the train of stimuli compared to control conditions. The blockade of Kir4.1 channels located on both oligodendrocytes and astrocytes (Olsen and Sontheimer, 2008; Larsen and MacAulay, 2014; Larson et al., 2018) interferes with the normal K^+ buffering, favouring depolarisation of cells and their discharge. The possibility that the divalent cation Ba²⁺, in addition to the effect on K^+ buffering, can permeate Ca²⁺-channels and that *per se* it induces neuronal depolarization, that in turn could enhance a further increase of extracellular potassium, should also be taken into

account to explain the spiking and the hyperexcitability in PC_{deep} observed in our experiments (Nelson, 1986; Sutor and Hablitz, 1993). It has been reported that barium blocks Kir4.1 channels at micromolar concentration (De Saint Jan and Westbrook, 2005; Djukic et al., 2007; Bay and Butt, 2012), whereas it nonselectively blocks other K^+ channels at higher concentration (1 mM), such as TREK-1 (Ma et al., 2011) or Ca²⁺-activated potassium channels (Vergara and Latorre, 1983), therefore in our experiment a selective blockade of Kir4.1 channels is expected. Barium has been used as permeant ion in studies on high voltage-activated calcium channels (HVACC) in PC layer II cells (Magistretti et al., 1999). The permeability of Ca²⁺ and Ba²⁺ is similar in low voltage-activated Ca²⁺ channels (McRory et al., 2001), whereas in HVACC it has been shown that when Ba²⁺ replaces Ca²⁺ as charge carrier in equimolar concentration, it elicits larger current than Ca²⁺ in equimolar concentration (Yar and Winlow, 2016). However, in those studies, barium is tested in millimolar concentration (> 2 mM).

Unfortunately, oligodendrocytes Kir4.1 channel blockers more selective than BaCl₂ are not currently available and the specific

contribution of oligodendrocytes could not be assessed in more detail.

5. Conclusions

Our results suggest that unmyelinated fibers in PC_{sup} represent a preferential site of accumulation of K⁺ in the extracellular space. In future experiments, the correlation between extracellular K⁺ changes and the simultaneously recorded extracellular and intracellular activities could add interesting information to evaluate the impact of [K⁺]_o fluctuations on neuronal membrane excitability, but this is beyond the scope of this study.

In healthy conditions, in unmyelinated PC_{sup} fast activity-dependent K⁺ increases are counterbalanced by fast K⁺ decay, whereas in PC_{deep} myelinated regions K⁺ dynamics are slower. From the observations here reported, we can expect that when a focal pathological alteration of myelin occurs, a greater excitability and seizures mediated by [K⁺]_o in the periaxonal space may occur, as recently proposed (de Curtis et al., 2021; see also Li et al., 2024). Damaged axonal myelin coating implies a less effective extracellular K⁺ buffering by oligodendrocytes, that results in high [K⁺]_o around the naked, damaged fibers, which in turn induces axonal depolarization and promotes neuronal firing and hyperexcitability (Kofuji and Newman, 2004; Wallraff et al., 2006). In demyelinated axons of rat spinal roots, ectopic spikes have been recorded (Baker and Bostock, 1992; Smith et al., 1997) and an increase of Na⁺ and K⁺ channels both mediating pathological ion fluxes has been reported in genetic mouse models of seizures (Noebels et al., 1991) and in acquired models of demyelination (Waxman et al., 2004; Bagchi et al., 2014).

Authors contributions

LU and GB carried out the experiments and analyzed the data. LU and MdC conceived, designed and supervised the study. LU, GB and MdC wrote the paper. All authors have read and approved the manuscript.

Funding

The study was funded by the European Union - Next Generation EU - NRRP M6C2 - Investment 2.1 Enhancement and strengthening of biomedical research in the NHS (PNRR-MAD-2022-12376068 grant). GB is presently supported by a fellowship of the University of Pavia Doctorate program in Biomedical Sciences.

CRediT authorship contribution statement

Laura Uva: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Gaia Bruno:** Writing – review & editing, Writing – original draft, Validation, Investigation. **Marco de Curtis:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

None.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

Avoli, M., 1996. GABA-mediated synchronous potentials and seizure generation. *Epilepsia* 37, 1035–1042. <https://doi.org/10.1111/J.1528-1157.1996.TB01022.X>.

- Avoli, M., Louvel, J., Kurcewicz, I., Pumain, R., Barbarosie, M., 1996. Extracellular free potassium and calcium during synchronous activity induced by 4-aminopyridine in the juvenile rat hippocampus. *J. Physiol.* 493, 707–717. <https://doi.org/10.1113/JPHYSIOL.1996.SP021416>.
- Bagchi, B., Al-Sabi, A., Kaza, S., Scholz, D., O'Leary, V.B., Dolly, J.O., Ovsepian, S.V., 2014. Disruption of myelin leads to ectopic expression of KV1.1 channels with abnormal conductivity of optic nerve axons in a cuprizone-induced model of demyelination. *PLoS One* 9, e87736. <https://doi.org/10.1371/journal.pone.0087736>.
- Baker, M., Bostock, H., 1992. Ectopic activity in demyelinated spinal root axons of the rat. *J. Physiol.* 451, 539–552. <https://doi.org/10.1113/jphysiol.1992.sp019178>.
- Bay, V., Butt, A.M., 2012. Relationship between glial potassium regulation and axon excitability: a role for glial Kir4.1 channels. *Glia* 60, 651–660. <https://doi.org/10.1002/glia.22299>.
- Beckner, M.E., 2020. A roadmap for potassium buffering/dispersion via the glial network of the CNS. *Neurochem. Int.* 136, 104727. <https://doi.org/10.1016/j.neuint.2020.104727>.
- Biella, G., de Curtis, M., 1995. Associative synaptic potentials in the piriform cortex of the isolated guinea-pig brain in vitro. *Eur. J. Neurosci.* 7, 54–64. [https://doi.org/10.1016/S0920-1211\(96\)00042-3](https://doi.org/10.1016/S0920-1211(96)00042-3).
- Biella, G., de Curtis, M., 2000. Olfactory inputs activate the medial entorhinal cortex via the hippocampus. *J. Neurophysiol.* 83, 1924–1931. <https://doi.org/10.1152/jn.2000.83.4.1924>.
- Butt, A.M., Ransom, B.R., 1989. Visualization of oligodendrocytes and astrocytes in the intact rat optic nerve by intracellular injection of lucifer yellow and horseradish peroxidase. *Glia* 2, 470–475. <https://doi.org/10.1002/GLIA.440020609>.
- Carriero, G., Uva, L., Gnatkovsky, V., De Curtis, M., 2009. Distribution of the olfactory fiber input into the olfactory tubercle of the in vitro isolated guinea pig brain. *J. Neurophysiol.* 101, 1613–1619. <https://doi.org/10.1152/jn.90792.2008>.
- Carriero, G., Uva, L., Gnatkovsky, V., Avoli, M., De Curtis, M., 2010. Independent epileptiform discharge patterns in the olfactory and limbic areas of the in vitro isolated guinea pig brain during 4-aminopyridine treatment. *J. Neurophysiol.* 103, 2728–2736. <https://doi.org/10.1152/jn.00862.2009>.
- Chaban, Y.H.G., Chen, Y., Hertz, E., Hertz, L., 2017. Severe convulsions and Dysmyelination in both Jimpy and Cx32/47 –/– mice may associate astrocytic L-channel function with myelination and Oligodendrocytic Connexins with Intermodal Kv channels. *Neurochem. Res.* 42, 1747–1766. <https://doi.org/10.1007/s11064-017-2194-z>.
- David, G., Barrett, J.N., Barrett, E.F., 1993. Activation of internodal potassium conductance in rat myelinated axons. *J. Physiol.* 472, 177–202. <https://doi.org/10.1113/JPHYSIOL.1993.SP019942>.
- de Curtis, M., Pare, D., Llinas, R.R., 1991. The electrophysiology of the olfactory-hippocampal circuit in the isolated and perfused adult mammalian brain in vitro. *Hippocampus* 1, 341–354. <https://doi.org/10.1002/hipo.450010402>.
- de Curtis, M., Biella, G., Buccellati, C., Folco, G., 1998. Simultaneous investigation of the neuronal and vascular compartments in the guinea pig brain isolated in vitro. *Brain Res. Protocol.* 3, 221–228. [https://doi.org/10.1016/S1385-299X\(98\)00044-0](https://doi.org/10.1016/S1385-299X(98)00044-0).
- de Curtis, M., Uva, L., Gnatkovsky, V., Librizzi, L., 2018. Potassium dynamics and seizures: why is potassium ictogenic? *Epilepsy Res.* 143, 50–59. <https://doi.org/10.1016/j.epilepsyres.2018.04.005>.
- de Curtis, M., Uva, L., Lévesque, M., Biella, G., Avoli, M., 2019. Piriform cortex ictogenicity in vitro. *Exp. Neurol.* 321, 113014. <https://doi.org/10.1016/j.expneurol.2019.113014>.
- de Curtis, M., Garbelli, R., Uva, L., 2021. A hypothesis for the role of axon demyelination in seizure generation. *Epilepsia* 62, 583–595. <https://doi.org/10.1111/epi.16824>.
- de Curtis, M., Librizzi, L., Uva, L., 2023. Limbic and olfactory cortical circuits in focal seizures. *Neurobiol. Dis.* 178, 106007. <https://doi.org/10.1016/j.nbd.2023.106007>.
- De Saint Jan, D., Westbrook, G.L., 2005. Detecting activity in olfactory bulb glomeruli with astrocyte recording. *J. Neurosci.* 25, 2917. <https://doi.org/10.1523/JNEUROSCI.5042-04.2005>.
- Devinsky, O., Vezzani, A., O'Brien, T.J., Jette, N., Scheffer, I.E., De Curtis, M., Perucca, P., 2018. Epilepsy. *Nat. Rev. Dis. Primers* 4, 18024. <https://doi.org/10.1038/nrdp.2018.24>.
- Djukic, B., Casper, K.B., Philpot, B.D., Chin, L.S., McCarthy, K.D., 2007. Conditional Knock-out of Kir4.1 leads to glial membrane depolarization, inhibition of potassium and glutamate uptake, and enhanced short-term synaptic potentiation. *J. Neurosci.* 27, 11354. <https://doi.org/10.1523/JNEUROSCI.0723-07.2007>.
- Fertziger, A.P., Ranck, J.B.J., 1970. Potassium accumulation in interstitial space during epileptiform seizures. *Exp. Neurol.* 26, 571–585. [https://doi.org/10.1016/0014-4886\(70\)90150-0](https://doi.org/10.1016/0014-4886(70)90150-0).
- Fröhlich, F., Bazhenov, M., Iragui-Madoz, V., Sejnowski, T.J., 2008. Potassium dynamics in the epileptic cortex: new insights on an old topic. *Neuroscientist* 14, 422–433. <https://doi.org/10.1177/1073858408317955>.
- Garbelli, R., Milesi, G., Medici, V., Villani, F., Didato, G., Deleo, F., D'Incerti, L., Morbin, M., Mazzoleni, G., Giovagnoli, A.R., Parente, A., Zucca, I., Mastropietro, A., Spreafico, R., 2012. Blurring in patients with temporal lobe epilepsy: clinical, high-field imaging and ultrastructural study. *Brain* 135, 2337–2349. <https://doi.org/10.1093/brain/awr149>.
- Giaume, C., Naus, C.C., Sáez, J.C., Leybaert, L., 2021. Glial Connexins and Pannexins in the healthy and diseased brain. *Physiol. Rev.* 101, 93–145. <https://doi.org/10.1152/physrev.00043.2018>.
- Grimaldi, M., Atzori, M., Ray, P., Alkon, D.L., 2001. Mobilization of calcium from intracellular stores, potentiation of neurotransmitter-induced calcium transients, and Capacitative calcium entry by 4-Aminopyridine. *J. Neurosci.* 21, 3135–3143. <https://doi.org/10.1523/JNEUROSCI.21-09-03135.2001>.

- Haberly, L.B., 2001. Parallel-distributed processing in olfactory cortex: new insights from morphological and physiological analysis of neuronal circuitry. *Chem. Senses* 26, 551–576. <https://doi.org/10.1093/CHEMSE/26.5.551>.
- Heinemann, U., Lux, H.D., Gutnick, M.J., 1977. Extracellular free calcium and potassium during paroxysmal activity in the cerebral cortex of the cat. *Exp. Brain Res.* 27, 237–243. <https://doi.org/10.1007/BF00235500>.
- Heinemann, U., Konnerth, A., Pumain, R., Wadman, W.J., 1986. Extracellular calcium and potassium concentration changes in chronic epileptic brain tissue. *Adv. Neurol.* 44, 641–661.
- Kettenmann, H., Ranson, B.R., 1988. Electrical coupling between astrocytes and between oligodendrocytes studied in mammalian cell cultures. *Glia* 1, 64–73. <https://doi.org/10.1002/GLIA.440010108>.
- Kofuji, P., Newman, E.A., 2004. Potassium buffering in the central nervous system. *Neurosci* 129, 1043–1054. <https://doi.org/10.1016/j.neuroscience.2004.06.008>.
- Larsen, B.R., MacAulay, N., 2014. Kir4.1-mediated spatial buffering of K(+): experimental challenges in determination of its temporal and quantitative contribution to K(+) clearance in the brain. *Channels (Austin)* 8, 544–550. <https://doi.org/10.4161/19336950.2014.970448>.
- Larsen, B.R., Stoica, A., MacAulay, N., 2016. Managing brain extracellular K+ during neuronal activity: the physiological role of the Na+/K+-ATPase subunit isoforms. *Front. Physiol.* 7, 141. <https://doi.org/10.3389/fphys.2016.00141>.
- Larson, V.A., Mironova, Y., Vanderpool, K.G., Waisman, A., Rash, J.E., Agarwal, A., Bergles, D.E., 2018. Oligodendrocytes control potassium accumulation in white matter and seizure susceptibility. *Elife* 7, e34829. <https://doi.org/10.7554/eLife.34829>.
- Li, J., Qi, H., Chen, Y., Zhu, X., 2024. Epilepsy and demyelination: towards a bidirectional relationship. *Prog. Neurobiol.* 234, 102588. <https://doi.org/10.1016/J.PNEUROBIO.2024.102588>.
- Librizzi, L., Biella, G., Cimino, C., de Curtis, M., 1999. Arterial supply of limbic structures in the guinea pig. *J. Comp. Neurol.* 411, 674–682. [https://doi.org/10.1002/\(sici\)1096-9861\(19990906\)411:4<674::aid-cne11>3.0.co;2-o](https://doi.org/10.1002/(sici)1096-9861(19990906)411:4<674::aid-cne11>3.0.co;2-o).
- Librizzi, L., Janigro, D., De Biasi, S., de Curtis, M., 2001. Blood brain barrier preservation in the in vitro isolated guinea-pig brain preparation. *J. Neurosci. Res.* 66, 289–297. <https://doi.org/10.1002/jps.23022>.
- Ma, X.Y., Yu, J.M., Zhang, S.Z., Liu, X.Y., Wu, B.H., Wei, X.L., Yan, J.Q., Sun, H.L., Yan, H.T., Zheng, J.Q., 2011. External Ba²⁺ block of the two-pore domain potassium channel TREK-1 defines conformational transition in its selectivity filter. *J. Biol. Chem.* 286, 39813–39822. <https://doi.org/10.1074/jbc.M111.264788>.
- Magistretti, J., Brevi, S., De Curtis, M., 1999. Biophysical and pharmacological diversity of high-voltage-activated calcium currents in layer II neurones of guinea-pig piriform cortex. *J. Physiol.* 518, 705–720. <https://doi.org/10.1111/J.1469-7793.1999.0705P.X>.
- Martin-Lopez, E., Ishiguro, K., Greer, C.A., 2019. The laminar Organization of Piriform Cortex Follows a selective developmental and migratory program established by cell lineage. *Cereb. Cortex* 29, 1–16. <https://doi.org/10.1093/cercor/bhx291>.
- McRory, J.E., Santi, C.M., Hamming, K.S.C., Mezeyova, J., Sutton, K.G., Baillie, D.L., Stea, A., Snutch, T.P., 2001. Molecular and functional characterization of a family of rat brain T-type calcium channels. *J. Biol. Chem.* 276, 3999–4011. <https://doi.org/10.1074/JBC.M008215200>.
- Moody, W.J., Futamachi, K.J., Prince, D.A., 1974. Extracellular potassium activity during epileptogenesis. *Exp. Neurol.* 42, 248–263. [https://doi.org/10.1016/0014-4886\(74\)90023-5](https://doi.org/10.1016/0014-4886(74)90023-5).
- Muhlethaler, M., de Curtis, M., Walton, K., Llinas, R., 1993. The isolated and perfused brain of the guinea-pig in vitro. *Eur. J. Neurosci.* 5, 915–926. <https://doi.org/10.1111/j.1460-9568.1993.tb00942.x>.
- Murata, K., Kinoshita, T., Ishikawa, T., Kuroda, K., Hoshi, M., Fukazawa, Y., 2020. Region- and neuronal-subtype-specific expression of Na,K-ATPase alpha and beta subunit isoforms in the mouse brain. *J. Comp. Neurol.* 528, 2654–2678. <https://doi.org/10.1002/CNE.24924>.
- Nelson, M.T., 1986. Interactions of divalent cations with single calcium channels from rat brain synaptosomes. *J. Gen. Physiol.* 87, 201–222. <https://doi.org/10.1085/JGP.87.2.201>.
- Neville, K.R., Haberly, L.B., 2004. Olfactory cortex, in: Shepherd G.M. (5th ed.), the synaptic Organization of the Brain. Oxford university press, New York, pp. 415–454.
- Newman, E.A., 1986. High potassium conductance in astrocyte endfeet. *Science* 233, 453–454. <https://doi.org/10.1126/SCIENCE.3726539>.
- Noebels, J.L., Marcom, P.K., Jalilian-Tehrani, M.H., 1991. Sodium channel density in hypomyelinated brain increased by myelin basic protein gene deletion. *Nature* 352, 431–434. <https://doi.org/10.1038/352431a0>.
- Olsen, M.L., Sontheimer, H., 2008. Functional implications for Kir4.1 channels in glial biology: from K⁺ buffering to cell differentiation. *J. Neurochem.* 107, 589–601. <https://doi.org/10.1111/j.1471-4159.2008.05615.x>.
- Paletzki, R.F., Gerfen, C.R., 2019. Basic neuroanatomical methods. *Curr. Protoc. Neurosci.* 90, e84. <https://doi.org/10.1002/CPNS.84>.
- Poopalasundaram, S., Knott, C., Shamotienko, O.G., Foran, P.G., Dolly, J.O., Ghiani, C.A., Gallo, V., Wilkin, G.P., 2000. Glial heterogeneity in expression of the inwardly rectifying K⁺ channel, Kir4.1, in adult rat CNS. *Glia* 30, 362–372. [https://doi.org/10.1002/\(SICI\)1098-1136\(200006\)30:4<362::AID-GLIA50>3.0.CO;2-4](https://doi.org/10.1002/(SICI)1098-1136(200006)30:4<362::AID-GLIA50>3.0.CO;2-4).
- Prince, D.A., Lux, H.D., Neher, E., 1973. Measurement of extracellular potassium activity in cat cortex. *Brain Res.* 50, 489–495. [https://doi.org/10.1016/0006-8993\(73\)90758-0](https://doi.org/10.1016/0006-8993(73)90758-0).
- Pumain, R., Heinemann, U., 1985. Stimulus- and amino acid-induced calcium and potassium changes in rat neocortex. *J. Neurophysiol.* 53, 1–16. <https://doi.org/10.1152/JN.1985.53.1.1>.
- Rash, J.E., 2010. Molecular disruptions of the panglial syncytium block potassium siphoning and axonal saltatory conduction: pertinence to neuromyelitis optica and other demyelinating diseases of the central nervous system. *Neurosci* 168, 982–1008. <https://doi.org/10.1016/j.neuroscience.2009.10.028>.
- Schirmer, L., Möbius, W., Zhao, C., Cruz-Herranz, A., Ben Haim, L., Cordano, C., Shiow, L.R., Kelley, K.W., Sadowski, B., Timmons, G., Pröbstel, A.K., Wright, J.N., Sin, J.H., Devereux, M., Morrison, D.E., Chang, S.M., Sabeur, K., Green, A.J., Nave, K.A., Franklin, R.J.M., Rowitch, D.H., 2018. Oligodendrocyte-encoded kir4.1 function is required for axonal integrity. *Elife* 7, e36428. <https://doi.org/10.7554/eLife.36428>.
- Schofield, C.N., 1990. Properties of K-currents in unmyelinated presynaptic axons of brain revealed by extracellular polarisation. *Brain Res.* 507, 121–128. [https://doi.org/10.1016/0006-8993\(90\)90530-0](https://doi.org/10.1016/0006-8993(90)90530-0).
- Smith, K.J., Feltz, P.A., Kapoor, R., 1997. Axonal Hyperexcitability: mechanisms and role in symptom production in demyelinating diseases. *Neuroscientist* 3, 237–246.
- Sutor, B., Hablitz, J.J., 1993. Influence of barium on rectification in rat neocortical neurons. *Neurosci. Lett.* 157, 62–66. [https://doi.org/10.1016/0304-3940\(93\)90643-y](https://doi.org/10.1016/0304-3940(93)90643-y).
- Suzuki, N., Bekkers, J.M., 2010. Distinctive classes of GABAergic interneurons provide layer-specific phasic inhibition in the anterior piriform cortex. *Cereb. Cortex* 20, 2971–2984. <https://doi.org/10.1093/CERCOR/BHQ046>.
- Tress, O., Maglione, M., May, D., Pivneva, T., Richter, N., Seyfarth, J., Binder, S., Zlomuzica, A., Seifert, G., Theis, M., Dere, E., Kettenmann, H., Willecke, K., 2012. Panglial gap junctional communication is essential for maintenance of myelin in the CNS. *J. Neurosci.* 32, 7499–7518. <https://doi.org/10.1523/JNEUROSCI.0392-12.2012>.
- Uva, L., De Curtis, M., 2005. Polysynaptic olfactory pathway to the ipsi- and contralateral entorhinal cortex mediated via the hippocampus. *Neurosci* 130, 249–258. <https://doi.org/10.1016/j.neuroscience.2004.08.042>.
- Uva, L., Trombin, F., Carriero, G., Avoli, M., De Curtis, M., 2013. Seizure-like discharges induced by 4-aminopyridine in the olfactory system of the in vitro isolated guinea pig brain. *Epilepsia* 54, 605–615. <https://doi.org/10.1111/epi.12133>.
- Uva, L., Saccucci, S., Chikhladze, M., Tassi, L., Gnatkovsky, V., Milesi, G., Morbin, M., de Curtis, M., 2017. A novel focal seizure pattern generated in superficial layers of the olfactory cortex. *J. Neurosci.* 37, 3544–3554. <https://doi.org/10.1523/JNEUROSCI.2239-16.2016>.
- Vergara, C., Latorre, R., 1983. Kinetics of Ca²⁺-activated K⁺ channels from rabbit muscle incorporated into planar bilayers: evidence for a Ca²⁺ and Ba²⁺ blockade. *J. Gen. Physiol.* 82, 543–568. <https://doi.org/10.1085/JGP.82.4.543>.
- Wallraff, A., Köhling, R., Heinemann, U., Theis, M., Willecke, K., Steinhäuser, C., 2006. The impact of astrocytic gap junctional coupling on potassium buffering in the hippocampus. *J. Neurosci.* 26, 5438–5447. <https://doi.org/10.1523/JNEUROSCI.0037-06.2006>.
- Wang, H., Kunkel, D.D., Martin, T.M., Schwartzkroin, P.A., Tempel, B.L., 1993. Heteromultimeric K⁺ channels in terminal and juxtaparanodal regions of neurons. *Nature* 365, 75–79. <https://doi.org/10.1038/365075a0>.
- Waxman, S.G., Craner, M.J., Black, J.A., 2004. Na⁺ channel expression along axons in multiple sclerosis and its models. *Trends Pharmacol. Sci.* 25, 584–591. <https://doi.org/10.1016/j.tips.2004.09.001>.
- Wenzel, M., Huberfeld, G., Grayden, D.B., de Curtis, M., Trevelyan, A.J., 2023. A debate on the neuronal origin of focal seizures. *Epilepsia* 64, S37–S48. <https://doi.org/10.1111/EPI.17650>.
- Yar, T., Winlow, W., 2016. Isolation and characterization of whole-cell Calcium Channel currents in Cultured. Identified Neurones of Lymnaea. *EC Neurology* 3 (5), 459–481.